# Infinite Descent

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## 1 Introduction

### 1.1 What is Infinite Descent

Infinite descent, also known as Fermat's method of descent, is a way of proof by contradiction. It uses the fact that there must exist a smallest number satisfying a statement. Proof by infinite descent would show that if the statement holds for a number, then it holds for a smaller one, and so on, meaning there could not be a smallest number. This contradiction allows you to prove that no number satisfies the statement.

### 1.2 Definitions, Theorems, and Corollaries

**Definition 1.1** (Rational Number)**.** Rational numbers are numbers that can be written as the ratio of two integers

**Theorem 1.1** (Well-Ordering Principle for natural numbers)**.** *Every non-empty set of natural numbers has a least element*

**Theorem 1.2** (Pythagorean Triple)**.** *If primitive ordered triple* $(x, y, z)$ *with* $gcd(x, y) = 1$ *satisfies* $x^2+y^2 = z^2$*, it is possible to write* $x = 2rs, y = r^2 - s^2, z = r^2 + s^2$ *with* $r$ *and* $s$ *being relatively prime integers.(The proof of this is left to the reader) For example,* $(3, 4, 5)$ *is a primitive Pythagorean triple, but* $(6, 8, 10)$ *is a Pythagorean triple, but not primitive.*

**Theorem 1.3** (Fermat's Last Theorem)**.** *There is no three positive integers a,b, and c satisfy the equation* $a^n + b^n = c^n$ *for any integer value of n¿2*

## 2 Some proofs that use infinite descent

### 2.1 Yeah, it is. So prove it.

$\sqrt{2}$ is irrational; this is true and much more intuitive to believe than to prove. You might have seen this proof that uses infinite descent. It goes as follows:

*Proof.* Assume that $\sqrt{2}$ is rational, that is, that there exist natural numbers $p$ and $q$ s.t. $\frac{p}{q} = \sqrt{2}$.

By the Well-Ordering Principle, there exists a smallest integer $p$ that satisfies the above equation. (or that there exist $p$ s.t. $\frac{p}{q}$ cannot be further simplified, equivalent to $\gcd(p, q) = 1$)

$\frac{p}{q} = \sqrt{2} \to \frac{p^2}{q^2} = 2 \to p^2 = 2q^2 \to 2 \mid p$

We can rewrite $p = 2p'$ with $p' \in \mathbb{N}$ By substituting $p = 2p'$, through a similar process, we can prove that $2 \mid q$. We will write $q = 2q'$ with $q' \in \mathbb{N}$ This means $\frac{p}{q} = \frac{2p'}{2q'} = \frac{p'}{q'}$ Notice how we can reduce $p$ infinitely, which means the set of possible values of $p$ doesn't have a smallest element. By infinite descent (or contradiction to Well-Ordering Principle), the assumption made is false, $\sqrt{2}$ is irrational.

□

## 2.2 FERMAT LAST THEOREM?! n=4.

Fermat, in the margin of his copy of the *Arithmetica*, wrote

> It is impossible to separate a cube into two cubes, or a fourth power into two fourth powers, or, in general, any power higher than the second, into two like powers. I have discovered a truly **marvelous** proof of this, which this margin is too narrow to contain.

Due to the limited space of the margin, Fermat's Last Theorem remained unsolved for more than 300 years until Andrew Wiles developed new techniques to prove it. Although this article cannot hope to contain Fermat's "marvelous proof," Fermat did succeed in proving the case where the exponent equals four, which goes as follows:

First, by proving that no ordered triple $(x, y, z)$ satisfies the equation $x^4 + y^4 = z^2$, it would immediately follow that $n = 4$ will not produce any solutions for FLT.

Now, by the Well-Ordering Principle, we can ensure the existence of $(x_0, y_0, z_0)$, with $z_0$ **being the smallest integer that satisfies the equation**, assuming that there exists at least one solution.

By Theorem 1.2, we can write

$$\begin{cases} x_0^2 = 2rs \\ y_0^2 = r^2 - s^2 \\ z_0 = r^2 + s^2 \end{cases} \tag{1}$$

Claim: s is even

Now we focus on the parity of $r$ and $s$; by the statement of Theorem 1.2, they cannot both be even or both be odd. Notice that if we make $r$ even and $s$ odd, $y_0^2 \equiv r^2 - s^2 \equiv 0 - 1 \equiv 3 \pmod 4$. And this cannot be true because no perfect square has a remainder of 3 when divided by 4. Therefore, we are left with the case where $s$ is even; our claim is proven.

Using this claim, we can write $s = 2p$, which led us to write $(\frac{x_0}{2})^2 = rp$. This result is especially useful, because since $r$ and $s$ are relatively prime, it is certain that $r$ and $p$ are relatively prime. Therefore, $r$ and $p$ must both be perfect squares, since their product is a perfect square. Thus, we will let $r = R^2$ and $p = P^2$.

By the second column of our definition of $y_0^2$, we know that $(2P^2)^2 + (y_0)^2 = (R^2)^2$. From here, we can find two relatively prime integers, $a$ and $b$, such that

$$\begin{cases} 2P^2 = 2ab \\ y_0 = a^2 - b^2 \\ R^2 = a^2 + b^2 \end{cases} \tag{2}$$

Using the result we have found in the previous paragraph, we can prove that $a$ and $b$ are perfect squares because $P^2 = ab$.

Now, we realize that there is a solution $(x, y, z) = (\sqrt{a}, \sqrt{b}, R)$ (why?). It is clear that $R < z_0$, and so an ordered triple with a smaller $z$ can be generated, which contradicts our claim that $z_0$ is the smallest integer that satisfies the equation, meaning that our assumption that there exist at least one ordered triple $(x, y, z)$ is false.

# 3 Vieta Jumping

## 3.1 Notorious Problem Six

### 3.1.1 Introduction

Ever since 1988, when the first problem using the method of Vieta Jumping appeared as the sixth problem on the IMO, Vieta Jumping has become well-known among problem solvers.

Vieta Jumping is another variant of Infinite Descent that uses Vieta's Formula to generate a smaller solution. The problem that uses this method goes as follows:

"Let $a$ and $b$ be positive integers such that $ab+1$ divides $a^2+b^2$. Show that $\frac{a^2+b^2}{ab+1}$ is the square of an integer." (Hint: Let $\frac{a^2+b^2}{ab+1} = c$. Think about how to relate $a, b, c$ and how many solutions of $a$ exist, given $b$ and $c$)

## 3.2 Group Law on Conics

**Theorem 3.1.** *For any real coefficient $c_1$ and $c_2$, there must be at least one real solution $x$ such that $c_1 \sin x + c_2 \cos x = 0$*

The proof will be left as an exercise for the reader, this lemma will come in handy for the proof of the more important theorem. (Hint: divide $\cos x$ on both sides, do consider the case where $\cos x = 0$ though)

**Theorem 3.2.** *Every conic section can be represented by an equation of the form $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$, where (A, B, C, D, E, F $\in \mathbb{Q}$). (we will restrict to rational conics) In particular, the mixed term (Bxy) is necessary to describe conics that are rotated with respect to the coordinate axes. Note that a conic with $B \neq 0$ is called an "oblique conic"*

*Proof.* We begin with an observation on how the coordinates of a point $(x_0, y_0)$ or in its polar form, $(r\cos\alpha, r\sin\alpha)$ can be expressed after rotating by $\beta$ (counterclockwise, of course), which is clearly $(r\cos(\alpha+\beta), r\sin(\alpha+\beta))$. By angle addition, that would imply that the "new" coordinates, which we denote as $x_1$ and $y_1$, will be:

$$\begin{cases} x_1 = r(\cos\alpha\cos\beta - \sin\alpha\sin\beta) \\ y_1 = r(\sin\alpha\cos\beta + \cos\alpha\sin\beta) \end{cases} \tag{3}$$

or even better:

$$\begin{cases} x_1 = x_0\cos\beta - y_0\sin\beta \\ y_1 = y_0\cos\beta + x_0\sin\beta \end{cases} \tag{4}$$

This is extremely useful because we can substitute that into the conic form, and hopefully, we can try to adjust $\beta$ so that the $xy$-term is canceled out. We should focus on the $x_0y_0$ term, which has a degree of two, meaning that in the substitution, only $Ax^2 + Bxy + Cy^2$ can contribute to the coefficient of the $x_0y_0$. Skipping the computations, after substitution, the coefficient turns out to be $(C-A)\sin 2\beta + B\cos 2\beta$. By Theorem 3.1, this clearly has at least one real $\beta$ such that the coefficient becomes zero, implying an oblique conic is just a rotation of a "standard" conic. □

**Theorem 3.3** (Vieta's Formula for quadratic)**.** *Let $x_1$ and $x_2$ be the roots of the quadratic $x^2 + ax + b$, it is true that $x_1 + x_2 = -a$ and $x_1x_2 = b$*

*Proof.* Clearly $(x - x_1)(x - x_2) = x^2 - (x_1 + x_2)x + x_1x_2 = x^2 + ax + b$. □

**Theorem 3.4** (Conjugate Root Theorem for Irrational Roots)**.** *If a polynomial with rational coefficient has an irrational root $A + \sqrt{B}$ then $A - \sqrt{B}$ must also be a root, or that irrational roots must come in pairs. ($A, B \in \mathbb{Q}$*

**Theorem 3.5** (Group Law on Conics)**.** *On a non-degenerate conic $C$, using a fixed rational point $N$ and two other rational points $A$ and $B$, a new rational point can be found, by drawing a line through $N$ parallel to $\overline{AB}$.*

*Proof.* The proof is quite elementary, as this can be reworded as follows: The intersection of a non-degenerate conic and a line with rational slope, over a rational point, is rational. Let the coordinates of point $N$ be $(x_N, y_N)$. The fact that $A$ and $B$ need to be rational implies the slope is rational (why?), which means this line can be expressed as $y = m(x - x_N) + y_N$, while the conic can be expressed as $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$. Substitution gives you a degree two equation, with one of the roots $N$ being rational and all of its coefficients being rational. By the Conjugate Root Theorem for Irrational Roots, the other intersection must also be a rational point; if not, then there isn't another root that can be its conjugate (A degree two polynomial has at most two distinct roots). □

This is the observation or theorem needed for the intuition of this problem.

### 3.3 The Solution

There are some other useful observations for this problem listed below:

- With $c$ being fixed, if $(a, b)$ is a solution, then $(b, a)$ is also a solution
- $a = b^3$ generates solutions where $c$ is a perfect square
- With $c$ being fixed, the relation between $a$ and $b$ happens to be a conic

Inspired by the group law of conics, it makes sense for us to first construct a diagram for an arbitrary perfect square $c$ value and start drawing lines between rational points (Note that the problem wants "lattice points" instead of rational points, but the rule also applies similarly). Notice how we can use points $A$, $A'$, and $B$ to construct more solutions. Focusing on points $B$ and $B'$, it's clear that they have the same $y$-coordinate. Plugging this $y$-value as $b$ produces two solutions for $a$, let's look more closely to it, take $(8, 2)$ as an example. $\frac{8^2+2^2}{8\times2+1} = 4$.
Let the $y$-value be replaced with the letter $y$ in this equation $\frac{y^2+8^2}{8y+1} = 4$ which is a quadratic with integer coefficients, with an integer root given, thus the other root must also be an integer.

*Proof.* Suppose $k$ is not a perfect square, let the lattice point $(A, B)$ lie on the conic produced by $\frac{a^2+b^2}{ab+1} = k$ which can be written as $a^2 - kb(a) + (b^2 - k) = 0$, such that $A$ is minimized. Without loss of generality, we can assume $A > B$.
Clearly, one of the roots of this quadratic is $a = A$. By Vieta's Formula for a quadratic, you can express the other root as $kb - a$ or $\frac{b^2-k}{a}$. Notice how this root has to be a positive integer, that is not zero ($b^2 \neq k$).

In addition, since $A > B$, $B^2 - k < B^2 < A^2$. This other root is also less than $A$. If $k$ is not a perfect square, there must exist a point $(x_2, B)$ less than $(A, B)$, the minimum element; thus, $k$ must be a perfect square. □

# 4 Conclusion

In conclusion, infinite descent is a powerful tool for solving Diophantine equations. The fundamental principle that you can disprove the existence of solutions by generating a smaller counterexample amazes me. While studying Vieta jumping, I began to wonder whether other number-theoretic tools could be used to solve some of the same problems, but I got nowhere. This process of asking "what else might work?" proved to be as valuable as learning the methods themselves. As a student, I truly wonder if there are other alternative solutions.

# 5 Reference

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